GE VERNOVA

GridOS SCADA

0.7.0

Tagging User Guide

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Change History

Date	Change Description
2025-Jun-24	GridOS SCADA 0.3.0: • This is a new document for GridOS SCADA 0.3.0.
2025-Aug-13	GridOS SCADA 0.4.0: • Added Tagging Configuration with information about configuration options related to the tagging-service in the SCADA system. • Added Tagging Permissions and AORs Validation to provide guidance on how user permissions and AOR validation work for tagging operations.

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1. Tagging Operations from Substation Tabular

> **Important**
>
> This user guide describes how operator can create, modify, and remove tags from the Substation Tabular display. The tagging side panel is currently available only for Points with Control. In future releases, it will be extended to other types of measurements and other types of displays, such as SLDs, schematics, and Geo-views.

1.1. Introduction

This guide provides step-by-step instructions for managing tags from the Substation Tabular display. Tagging enables you, as an operator, to manage tags and document the operational status of devices and associated measurements within the power system.

These instructions are intended for users with access to the GridOS interface and the necessary permissions to manage tags.

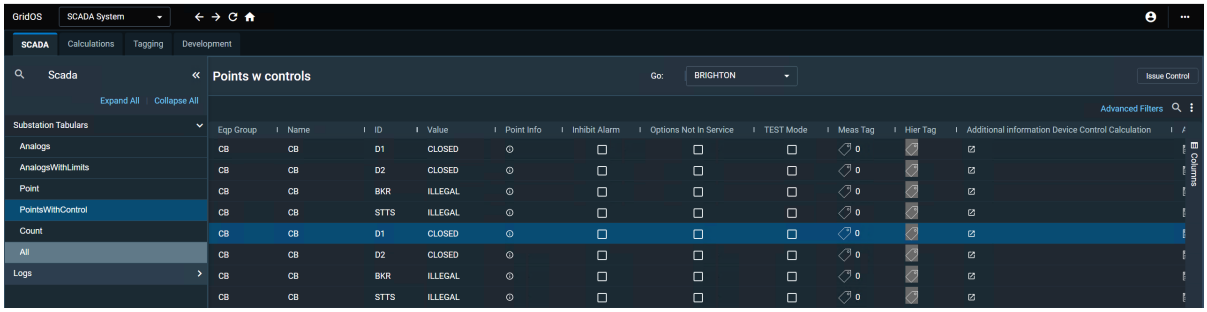
> **Note**
>
> This functionality is only available for a Hybrid solution consisting of GridOS and e-terra SCADA. In particular this is to address the needs of ADMS V1.

1.2. Creating a Tag

To create a new tag:

1. Navigate to the **Substation Tabular** display.

   Select **Points with Control** tab.

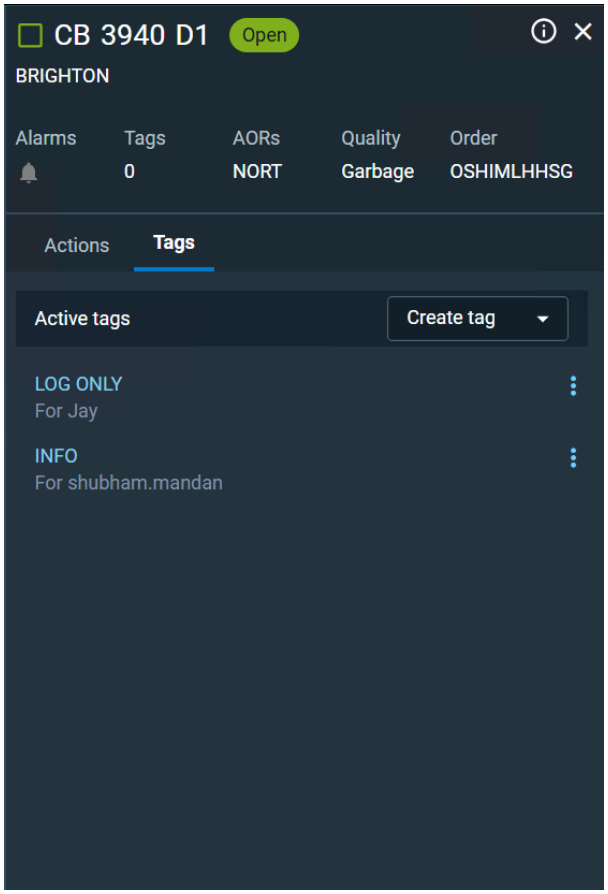


> **Note**
>
> The tagging side panel is currently available only for Points W Control. In future releases, it will be available for other types of measurements.

2. Select the desired substation from the dropdown list. A tabular view of the substation's points appears, with various columns, including "Meas Tag" for tagging purposes.
3. Locate the "Meas Tag" column and click the corresponding icon for the point you want to tag.

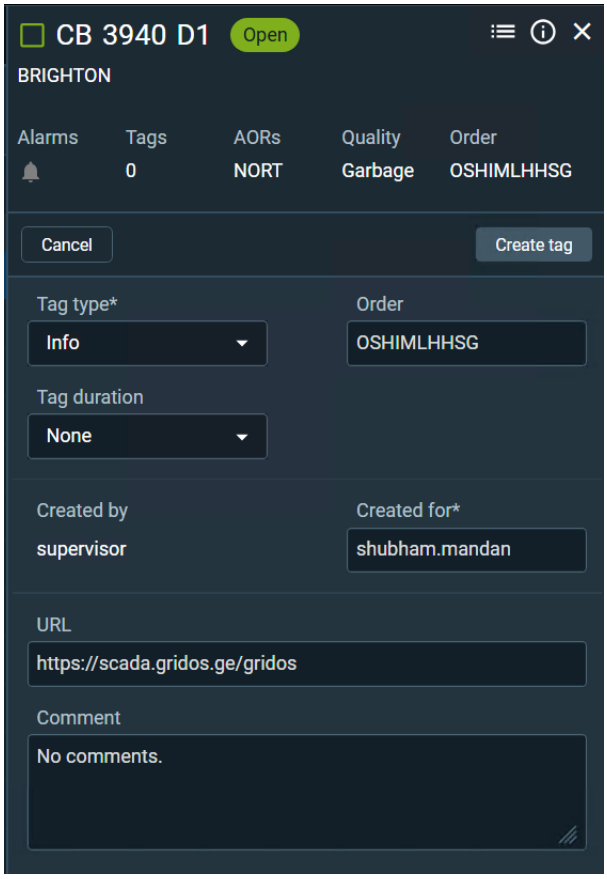
A side panel appears, showing point details and available actions. The fields in the side panel include:

- **Alarm Indication**: Shows the current alarm status for the point. The icon is yellow if any alarms are active; otherwise, it is grey.
- **Number of Tags**: Displays the count of tags applied to the point, indicating operational restrictions or notes.
- **Area**: Indicates the operational area or zone for the point. Only one area is shown currently; multiple areas will be supported in a future release.
- **Quality**: Reflects the current data quality or status (for example, Good, Bad, Uncertain). Only one value is shown currently; multiple values will be supported in a future release.
- **Order ID**: Displays the switching order identifier for the control operation. This field is not editable and is currently populated with mock data. In a future release, it will be sourced from the Switching Application.



4. Click on the **Create Tag** dropdown and select the appropriate tag type from the list.

A new side panel opens, presenting fields required to create a tag.



5. Complete the following fields:

> **Note**
>
> '*' next to field name indicates that the field is mandatory.

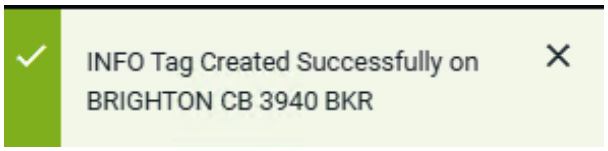
- **Tag Type**: Select the type of tag. Only applicable tag types (for example, Do Not Operate, Log Only) are shown based on system configuration and point eligibility.
- **Order**: Enter the order ID for the tag. This is typically a reference to a work order or operational instruction.
- **Tag Duration**: Specify how long the tag should remain active. Pick a value from the dropdown list.
- **Created by**: The user creating the tag. This field is auto-populated with your username and is not editable.
- **Created For**: Enter the username for whom the tag is being created. This may be another

user or group.

- **URL** (Optional): Provide a valid URL if applicable. Only valid URLs are accepted; invalid entries will be rejected.
- **Comment** (Optional): Add any relevant comments or notes regarding the purpose or context of the tag.

6. After filling in all required fields, click the **Create Tag** button.

   If the tag is created successfully, a confirmation message appears.



7. Return to the previous screen to verify that the new tag is listed under the active tags for the selected point.

> **Note**
>
> Creating tag and removing tag described here are performed from the Substation Tabular interface within GridOS, but the tags themselves are managed on the Heritage System.

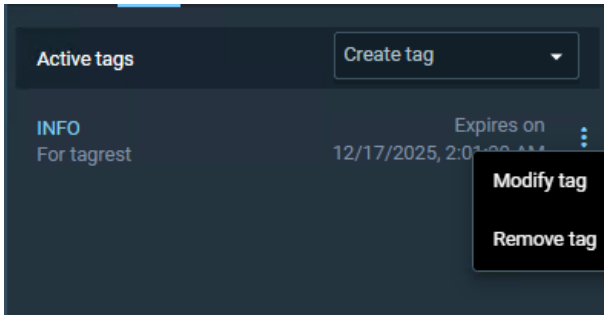
1.3. Removing a Tag

To remove an existing tag:

1. Navigate to the same side panel for the relevant point in the **Substation Tabular** display.

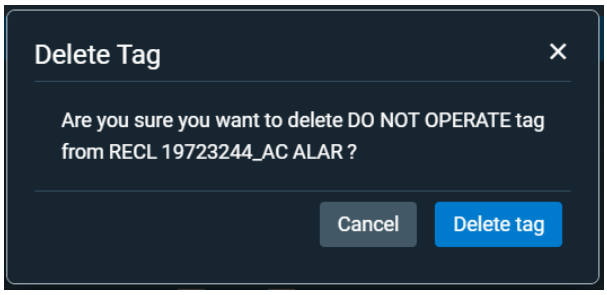
   A list of tags appears under the **Active tags** header.

2. Click the menu icon next to the tag you want to remove.
3. Select the **Remove tag** option from the menu.

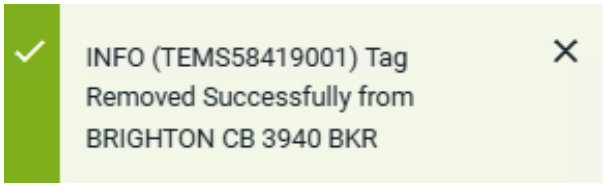


   A confirmation dialog appears.

4. Click **Remove tag** to confirm and remove the tag, or click **Cancel** to abort the operation.



If the tag is removed successfully, a confirmation message appears.



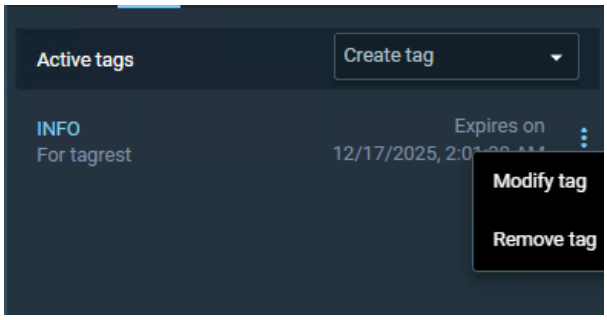
1.4. Modifying a Tag

To modify an existing tag:

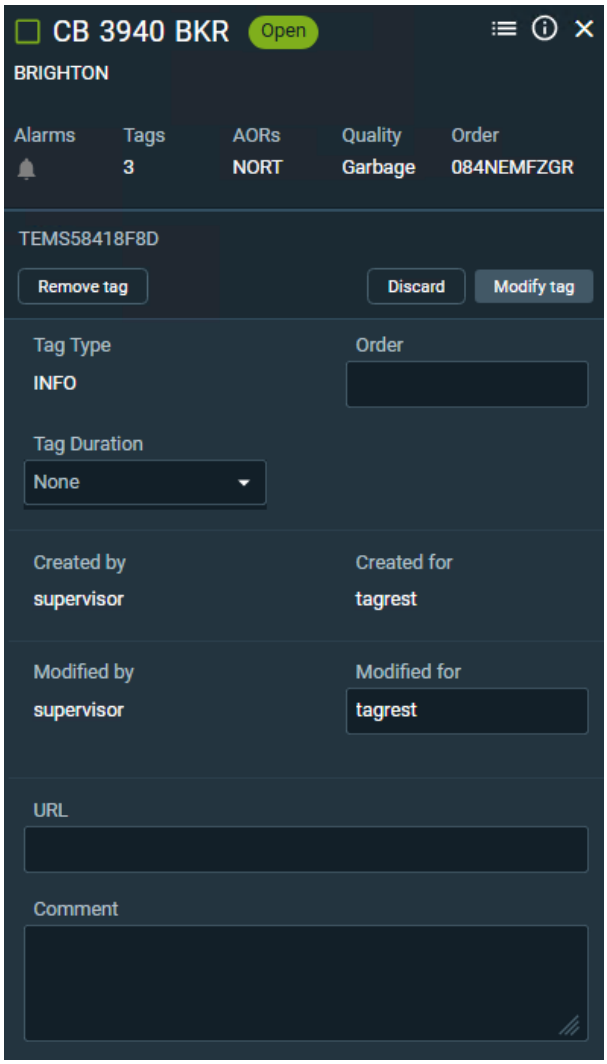
1. Navigate to the same side panel for the relevant point in the **Substation Tabular** display.

   A list of tags appears under the **Active tag** header.

2. Click the menu icon next to the tag you want to modify.
3. Select the **Modify tag** option from the menu.



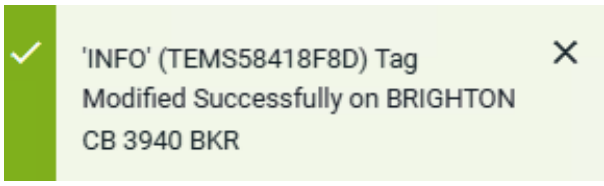
A new side panel opens, allowing you to modify the tag details.



4. Modify the following fields as needed:
   - **Tag Type**: The type of tag. This field is non-editable during tag modifications and displays the current tag type.
   - **Order**: Modify the order ID for the tag if necessary.
   - **Tag Duration**: Adjust the duration for which the tag should remain active.
   - **Created by**: The original creator of the tag. This field is non-editable and auto-populated.
   - **Created For**: The original user for whom the tag was created. This field is non-editable.
   - **Modified by**: The user/operator making the modification. This is auto-populated with your username.
   - **Modified for**: Enter the username for whom you are modifying the tag, if applicable.
   - **URL** (Optional): Modify the URL if needed. Only valid URLs are accepted.
   - **Comment** (Optional): Add or modify comments related to the tag modification.
5. After making the necessary changes, save or confirm the modifications as per the interface

instructions.

If the tag is modified successfully, a confirmation message appears.



1.5. Tagging Limitations and Constraints

- Only one tag can be created at a time for a given point.
- Tag groups cannot be created through this interface.
- Creating, modifying, and removing tags are subject to user permissions, and assigned AORs.
- Tags are managed on the Heritage system and may be subject to synchronization delays before appearing in GridOS displays.

1.6. Troubleshooting and Error Messages

- If a tag creation or update fails, ensure all required fields are completed and that the URL is valid.
- Only applicable tag types are shown; if a type is missing, check system configuration.
- If tags do not appear immediately after creation, allow time for synchronization.

2. Tagging Logging and Event Flow

This section explains how the GridOS Tagging logs and tracks all Tagging operations (create, edit, and delete). The system records every action with detailed context and outcome, and generates an event for audit and monitoring.

User can access logging and eventing for all Tagging, if you have the necessary AORs.

This section also explains how the system extracts event properties, handles permission errors, posts events to the Alarm Router Service, the tagging logs and where you can view the applicable logs and events.

2.1. Introduction

GridOS Tagging automatically logs all Tagging activities and generates events, which the system posts to the Alarm Router Service for traceability and compliance. As an Operator or Administrator, you can use these logs and events to audit Tagging actions, investigate issues, and ensure operational accountability.

2.2. What Gets Logged and Why

The following information is captured/logged for every Tagging operation (create, edit, delete):

- The type of operation (create, edit, delete)
- The tag type or tag ID
- The name of equipment tagged
- The user and the "device" (console, pc, tablet, etc.) from where the tagging operation is initiated

Log messages use templates defined in `messages/scada-plugin.json`:

ID	Severity	Format	Usage
SOP003	INFO	Received '%s' (%s) tag %s request for '%s' from %s by %s	When the system receives a tag edit/delete operation request.
SOP004	INFO	%s' (%s) tag %s successfully for '%s' from %s by %s	When the system completes a tag operation successfully.
SOP005	INFO	Received '%s' tag %s request for '%s' from %s by %s	When the system receives a tag create operation request.
SOP803	ERROR	Failed to %s '%s' (%s) for '%s' from %s by %s: %s	When a tag edit/delete operation fails (including permission errors).
SOP804	ERROR	Failed to %s '%s' for '%s' from %s by %s: %s	When a tag create operation fails (including permission errors).

2.2.1. How the System Uses Properties

- **Logging:** The system uses these properties as parameters for the log messages (SOP003, SOP004, SOP005, SOP803, SOP804), ensuring that each log entry contains full context about the operation.
- **Event Generation:** When generating a `GridAlarmEvent`, the system includes these properties in the event message and metadata, providing traceability for downstream systems.

2.2.2. Example

For a create tag request, the system might set the following properties:

Property	Value
tagTypeOrId	"Clearance"
equipmentName	"Main Transformer"
equipmentId	"123e4567-e89b-12d3-a456-426614174000"
action	"create"
createBy	"jdoe"
console	"ConsoleA"

The system uses these values to generate a log message like:

```
Received 'DO NOT OPERATE' tag create request for 'BRIGHTON CB 4143 BKR' from Organizer by supervisor
```

and includes them in the event sent to the Alarm Router Service.

2.3. Permission Handling

If a user attempts a tag operation without sufficient permissions, it will be denied. The following takes place:

- Logs SOP804 with the message "Insufficient Permission for Tagging" and all relevant properties
- Returns an error response to the user
- Posts an error event to the Alarm Router Service

Example log message for permission error

```
Failed to create 'TAG TESTMODE' for 'BRIGHTON CB 3940 D2' from Organizer by reader1: Insufficient
Permission for Tagging
```

2.4. Event Generation and Posting

After each operation (success or failure), the system prepares and posts an event to the Alarm Router Service. Each event includes:

- Event message (log message, e.g., SOP003, SOP004, SOP005, SOP803, SOP804)
- Equipment UUID
- Priority, genesis, and context (from configuration)
- Source time (current time)
- Data and event metadata

The event is serialized to JSON and posted to:

```
http://${env:ALARM_ROUTER_SERVICE_HOST}/api/v1/alarms
```

2.5. Viewing Logs and Events

You can view log messages and events in these locations:

2.5.1. Tagging Logs

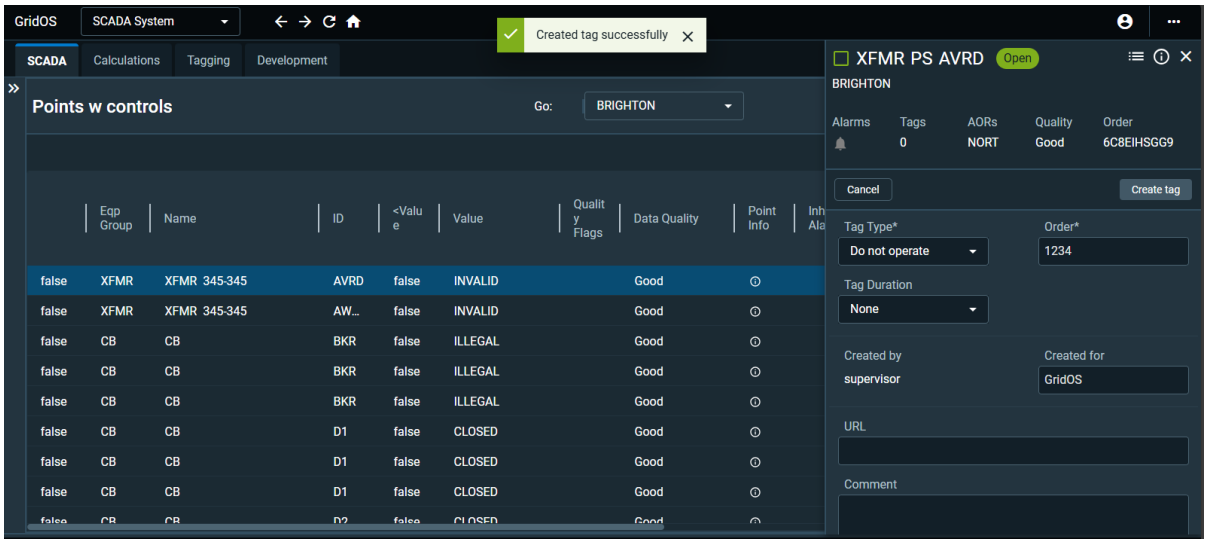
- All log messages (SOP003, SOP004, SOP005, SOP803, SOP804) are written to the application logs
- Access logs via Tagging application logs.

2.5.2. Alarm Router Service

- Tagging events (including errors and permission denials) are posted to the Alarm Router Service
- Events can be forwarded, stored, or displayed in:
  - Monitoring dashboards (e.g., Grafana, Kibana, Splunk), if the customer has these capabilities avaiable and configured (these are not part of GridOS SCADA)
  - Alarm/event management UIs
  - Notification systems (for example, the alert manager via GridOS Foundation, if configured)

2.6. Example Logging Flow

1. A user requests a Tagging operation (e.g., create).



2. The system extracts relevant properties (tag type or ID, action, equipment, user, console).
3. The system logs SOP005 for request receipt, SOP004 for success, or SOP804 for errors.
4. The system generates and posts a corresponding event to the Alarm Router Service.

Example success log

```
'INFO' (TEMS57714084) tag created successfully for 'BRIGHTON PS Primary PS AVRD' from Organizer by
supervisor
```

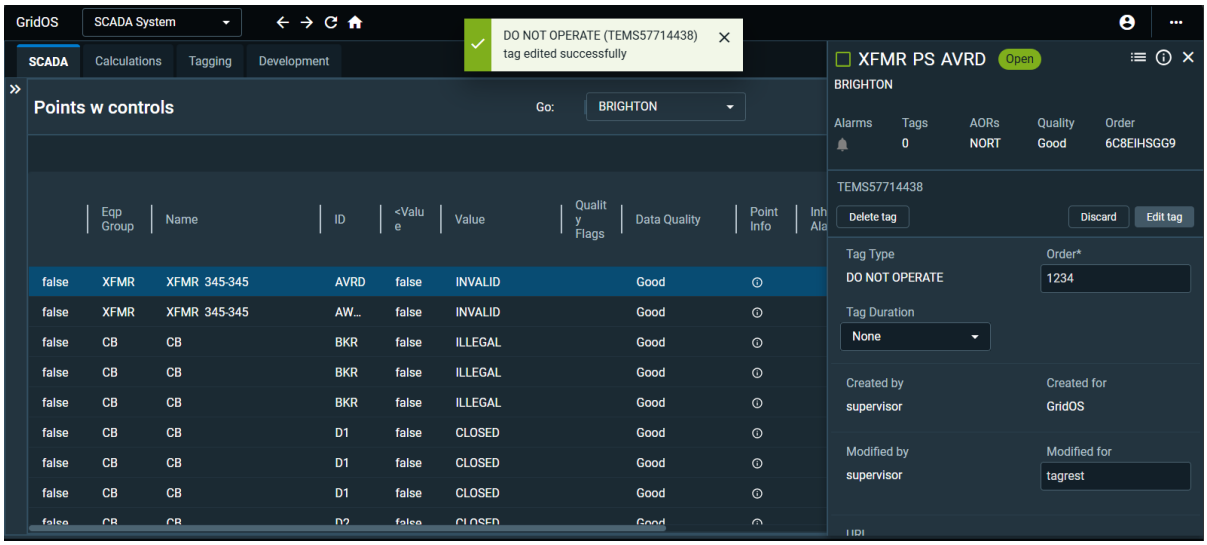
Example error log

```
Failed to create 'TAG TESTMODE' for 'BRIGHTON CB 3940 D2' from Organizer by reader1: Insufficient
Permission for Tagging
```

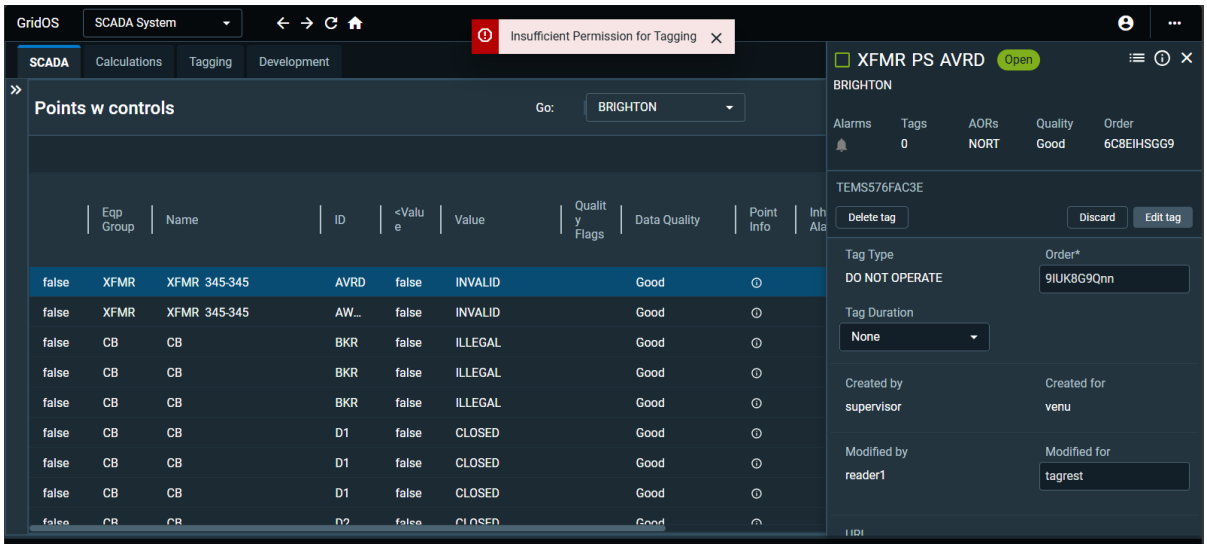
2.7. Tag Edit Request Example

Similarly, the following images show tag edit requests.

- Success case, when a user logs in with permissions for tag operations.



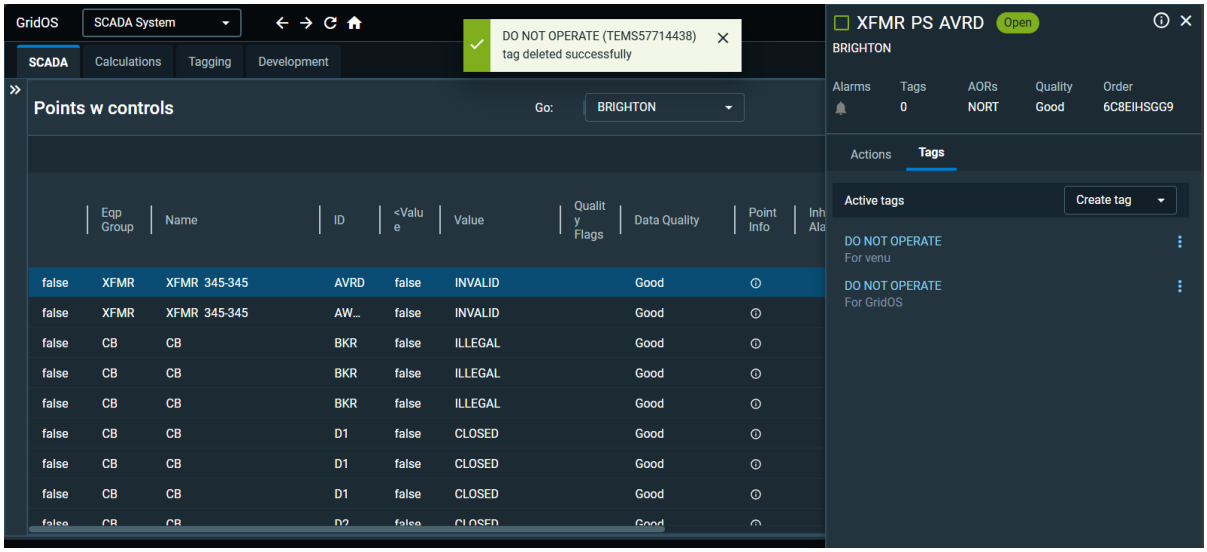
- Failure case, when a user doesn't have permissions for tag operations.



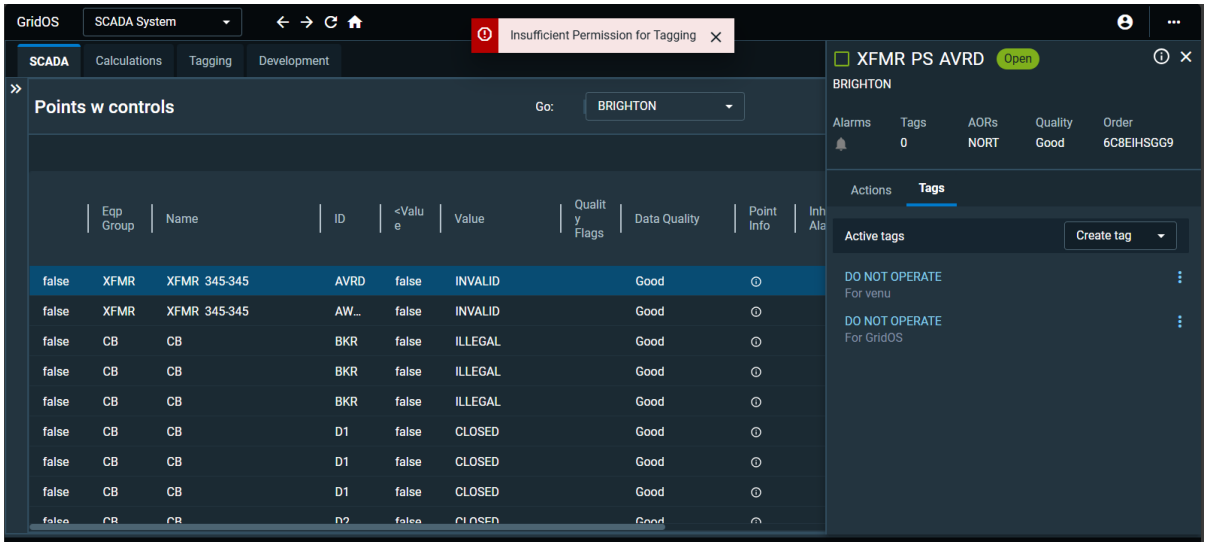
2.8. Tag Delete Request Examples

The following images show tag delete requests.

- Success case, when a user logs in with permissions for tag operations.

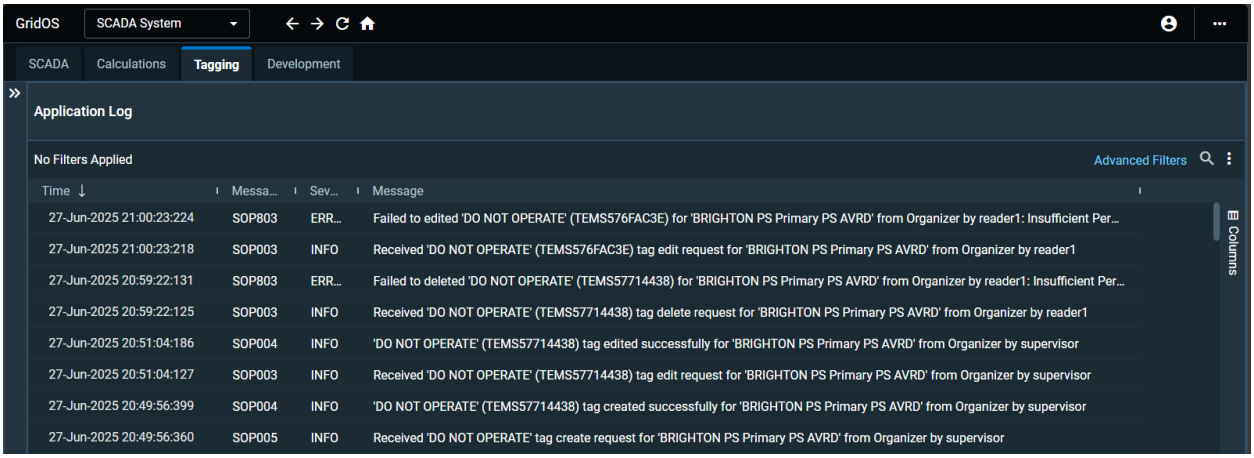


- Failure case, when a user doesn't have permissions for tag operations.



2.9. Tagging Application Log

Application log showing Tagging operation(s) messages:



2.10. Summary

- All Tagging operations are logged with full context and outcome.
- Events are generated for every operation and posted to the Alarm Router Service.
- Logs and events are available in plugin logs, the Alarm Router Service, and downstream monitoring systems.

2.11. References

- `TaggingRoutes.java`: src/main/java/com/ge/grid/organizer/plugin/scada/routes/TaggingRoutes.java
- `messages/scada-plugin.json`: src/main/resources/messages/scada-plugin.json

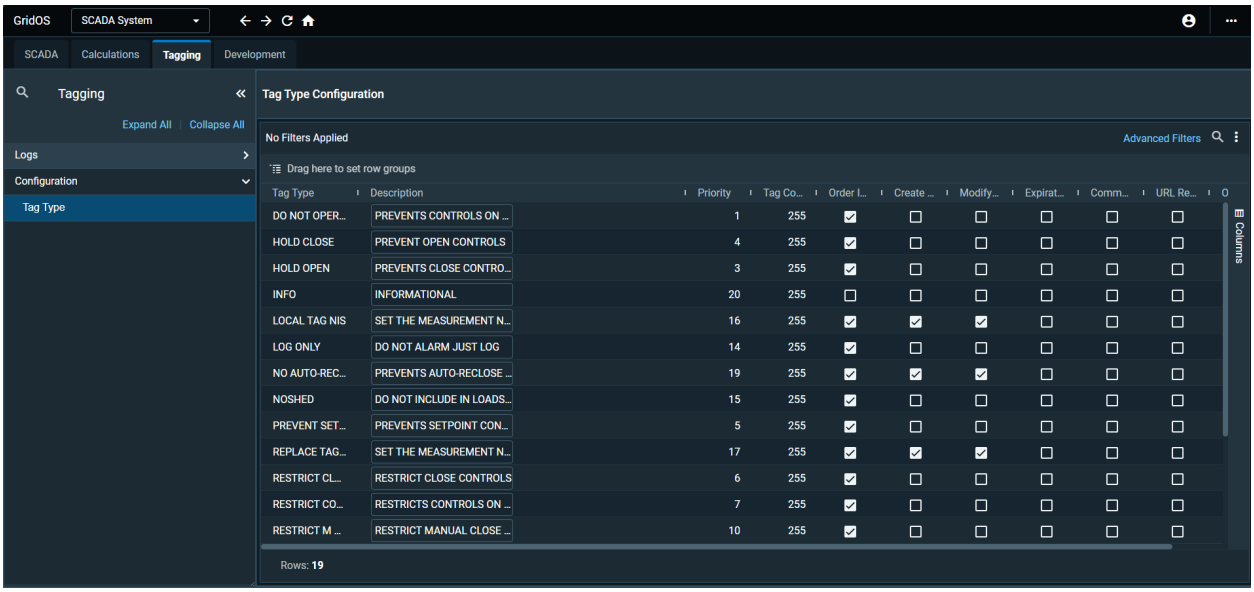
3. Tagging Configuration

3.1. Introduction

The Tagging Configuration UI provides a place for managing configuration options related to the tagging-service in the GridOS SCADA system. It allows administrators to control behaviour of tagging operations, including attribute options and restrictions, based on operational requirements.

3.2. Tag Type Configuration

This chapter covers modifying Tag Type attributes. This task is performed from the Tag Type Configuration display.



Only authorized users with the appropriate permissions can modify tag type configurations.

> **Note**
>
> If you do not see the Configuration tab or Tag Type options, ensure you have the necessary permissions or contact your system administrator.

3.2.1. Modifying a Tag Type Field

To modify a tag type in the Tag Type configuration table:

1. In the table, locate the tag type that you want to modify.

2. Use the available checkboxes to enable or disable attributes or set restrictions.

Refer to the following section for details about the fields that may be modified. Changes are applied immediately when a checkbox is clicked.

3.2.1.1. Modifiable Fields in Tag Type Configuration

The following fields are modifiable for each tag type:

Field	Description
tagTypeDescription	Description of the tag type
tagTypePriority	Priority of tags of this type (1 is highest)
tagTypeColor	RGB color value for tags of this type
tagTypeCharacter	Single character representing the tag type
blockOpen	Blocks open controls for this tag type
blockClose	Blocks close controls for this tag type
blockSetPoint	Blocks set point controls for this tag type
blockAutoReclose	Blocks automatic reclose controls for this tag type
noShed	Prevents load shedding for this tag type
earthLock	Locks the plant item in its earthed position
ivvcBlock	Inhibits IVVC from controlling the plant
restrictOpen	Restricts open controls (override possible with warning)
restrictClose	Restricts close controls (override possible with warning)
restrictSetPoint	Restricts set point controls (override possible with warning)
restrictManualOpen	Restricts manual open controls (override possible with warning)
restrictManualClose	Restricts manual close controls (override possible with warning)
allowExpiration	Allows expiration for tags of this type
idRequired	Requires order number or user ID for tags of this type
createForRequired	Requires createdFor for tag creation
modifyForRequired	Requires modifiedFor for tag modification
removeForRequired	Requires removedFor for tag deletion
commentRequired	Requires a comment for the tag
urlRequired	Requires a URL for the tag
tagGroupRequired	Requires a tag group for the tag
expirationRequired	Requires expiration for the tag
createForFill	Automatically fills createdFor with createdBy if not provided
modifyForFill	Automatically fills modifiedFor with modifiedBy if not provided
removeForFill	Automatically fills removedFor with removedBy if not provided
requireOpen	Requires the device to be in the opened state

Field	Description
requireClosed	Requires the device to be in the closed state
inheritTagGroupDetails	Inherits tag group details
onlyRemoveByCreator	Only the creator can remove tags of this type
tagGroupExclusive	Tag type is mutually exclusive within tag groups

Note

The current display of fields is temporary. Some fields will remain fixed as per modelling, while others may be modifiable by the administrator. This will be updated once categorization is finalized.

4. Tagging Permissions and Areas of Responsibility

This chapter explains how permissions and Area of Responsibility (AOR) validation work for tagging operations in the SCADA Organizer Plugin, including user roles, validation flow, and error handling.

4.1. Overview

Tagging operations are tightly integrated with permissions and Area of Responsibility (AOR) controls.

- **Permissions**: Restricts which users can perform tagging actions (create, edit, delete).
- **Area of Responsibility (AOR)**: Restricts actions to specific equipment or operational areas.

Permissions and AORs together ensure that tagging actions can be done only by users authorized for that action and only on devices or operational areas that the users are authorized for.

4.2. How Tagging Validation Works

4.2.1. Permissions Check

Permissions are determined by the following user groups:

- **Writer Group**: View tags, as well as create, edit, and delete tags.
- **Reader Group**: Can only view tags and cannot make changes.

When a user logs into the ADMS/AEMS UI, the system first checks the user group membership. If you do not have the required permissions (for example, a Reader trying to create a tag), the action is blocked, a message is displayed to the user, and an error event is posted to the Alarm Service.

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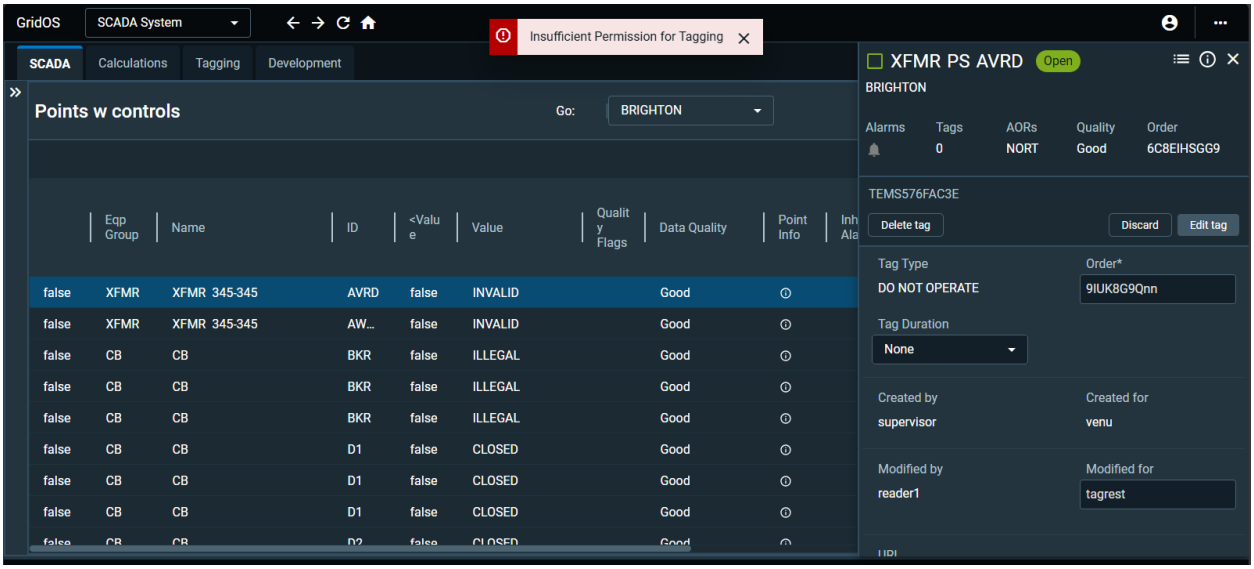

4.2.2. AOR Validation

For every tagging action, the Tagging Application verifies whether the logged-in user is assigned the AOR(s) associated with the equipment to which the tagging action is to be applied. The session includes a bearer token and client ID, which uniquely identify you. The system forms a URL using the client ID and calls an internal endpoint to fetch all AORs associated with the session.

When a tagging action is attempted, the requested AORs are compared with the list fetched for the session. If the session includes the required AORs, the action is allowed; otherwise, it is denied and a message explains the restriction.

AORs are validated when a user attempts to create, edit, or delete a tag. Tags may be viewed without AOR validation.

For more details on error messages, see the Tagging Logging and Event Flow.

4.3. References

- Tagging Operations from Substation Tabular
- Tagging Logging and Event Flow